



Mercury Exposure in Humboldt (*Spheniscus humboldti*) and Chinstrap (*Pygoscelis antarcticus*) Penguins Throughout the Chilean Coast and Antarctica

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Abstract

Penguins are reliable sentinels for environmental assessments of mercury (Hg) due to their longevity, abundance, high trophic level, and relatively small foraging areas. We analyzed Hg concentrations from blood and feathers of adult Humboldt penguins (*Spheniscus humboldti*) and feathers of chinstrap penguins (*Pygoscelis antarcticus*) from different reproductive colonies with variable degrees of urbanization and industrialization along the Chilean and Antarctic coasts. We evaluated Hg concentration differences between species, sexes (Humboldt penguins), and localities. Our results showed significantly greater levels in Humboldt penguins than in chinstrap penguins and nonsignificant differences between sexes among Humboldts. Penguin Hg concentrations showed a latitudinal pattern, with greater values of the metal at lower latitudes, independent of the species. Both studied penguin species showed elevated Hg concentrations compared to their congeners, highlighting the necessity to investigate potential negative effects on their populations. Although differences between species are possibly due to variation in diet and trophic level, our results suggest an important effect of the degree of Hg pollution adjacent to foraging areas. Further research on Hg content in prey species and environmental samples, together with a larger overall sample size, and investigation on penguin's diet and trophic level are needed to elucidate Hg bioavailability in each location and the role of local Hg pollution levels. Likewise, it is important to monitor Hg and other heavy metals of ecotoxicological importance in penguin populations in vulnerable regions of Chile.

Mercury (Hg) is a persistent heavy metal that is naturally found in the atmosphere at trace levels (Gochfeld 2003). Particularly since the industrial age, anthropogenic emissions have significantly increased in concentration, mainly due to industrial activities (e.g., mining, electricity, and

paper-making), wastewater disposal, and fossil fuel use (Gochfeld 2003; dos Santos et al. 2006). Mercury is a highly toxic, nonessential element of global concern, whose inorganic form is highly volatile and can be transported and deposited in the open ocean (Carravieri et al. 2016). In its organic form, Hg bioaccumulates and biomagnifies in marine food chains, resulting in elevated concentrations in

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tissues of higher trophic-level organisms (Gochfeld 2003; Frias et al. 2012; Blévin et al. 2013; Carravieri et al. 2016).

Given that Hg emissions are mainly anthropogenic in nature, the implications of mercury deposition in natural environments are based heavily on regional industrial activities. Chile is a country whose economy is strongly based on mining, with most mineral deposits (metallic and nonmetallic) located in the north, between 17 and 32°S (Castro and Sánchez 2003; Higuera et al. 2004). Heavy metals are a natural component of marine systems but also are released through human activities (Universidad de Chile 2016). The primary depositions of Hg are found between 26 and 32°S, where gold production is concentrated (Higuera et al. 2004). This activity traditionally used metallic mercury to extract gold through amalgamation (Castro and Sánchez 2003; Higuera et al. 2004), and this type of mining is environmentally destructive, contaminating groundwater and soil and releasing substantial quantities of Hg (Castro and Sánchez 2003; Leiva and Morales 2013). In fact, elevated Hg levels have been found in sediments, coastal waters, and benthic fauna of the area (Castilla and Nealler 1978; Universidad de Chile 2016). Much further south, Antarctica is increasingly being affected by anthropogenic activities (Espejo et al. 2014). Although most chemical levels in Antarctic ecosystems are very low, ongoing contamination could be affecting marine environments and Antarctica's endemic species (Espejo et al. 2014).

Birds are considered among the most reliable indicators of environmental changes (Robinson et al. 2012); seabirds constitute suitable subjects for mercury assessment because of their longevity and high trophic level, leading to high concentrations of metals in their tissues (Bearhop et al. 2000; Gochfeld and Burger 2001; Ochoa-Acuña et al. 2002; Blévin et al. 2013; Carvalho et al. 2013; Brasso et al. 2015). This combination of factors has led to the wide use of seabirds as mercury pollution indicators in different environments and locations (Thompson et al. 1998; Bearhop et al. 2000; Carravieri et al. 2013; Brasso et al. 2015; Finger et al. 2017).

Among seabirds, especially threatened species, feathers and blood have been frequently used for monitoring Hg pollution, because their collection is less invasive (Thompson et al. 1998; Becker et al. 2002; dos Santos et al. 2006). Feathers are the primary mercury excretion route in birds, containing approximately 70–93% of the total body burden (Evers et al. 1998) and reflecting metal input and storage between molts (Ochoa-Acuña et al. 2002). Blood, unlike feathers, reflects short-term exposure to contaminants and can be used to elucidate metal concentrations at small temporal and spatial scales (Thompson et al. 1998; Finger et al. 2015, 2016). Adverse effects of Hg in birds are associated to alterations in behavior and reproduction, such as lower hatchability, lower clutch and egg size, and increased nestling mortality (Carvalho et al. 2013).

Several seabird studies describe intraspecific variation of Hg concentrations; greater concentrations are observed in adults compared with chicks or juveniles (Becker et al. 2002; Frias et al. 2012; Carravieri et al. 2016). Differences also have been reported between sexes, which are thought to be a result of physiological or foraging strategy differences (Becker et al. 2002; Frias et al. 2012; Robinson et al. 2012). This context gave rise to the “egg depuration hypothesis,” in which egg-laying may allow Hg excretion; thus, males would have greater Hg concentrations in their tissues than females (Becker et al. 2002; Frias et al. 2012; Robinson et al. 2012; Polito et al. 2016). However, inter-specific variation also has been reported among several seabird species, which may be due to trophic position and food preference (Scheifler et al. 2005; Frias et al. 2012; Brasso et al. 2012; Carravieri et al. 2013; Brasso and Polito 2013).

The Humboldt penguin (*Spheniscus humboldti*) is one of four species of the genus *Spheniscus*, of which three inhabit South America. Its distribution range covers roughly 4500 km of the Eastern South Pacific coast, extending from Isla Foca in Peru (5°S) to Isla Guafo in southern Chile (43°S) (Birdlife International 2016a). This species is predominantly piscivorous, feeding on schooling fishes, including garfish (*Scomberesox saurus*), anchovy (*Engraulis ringens*), and Inca scad (*Trachurus murphyi*) (Herling et al. 2005). The Humboldt penguin has been classified as Vulnerable by the International Union for Conservation of Nature (IUCN), with populations remaining at persistently low levels since the early 1980s (Birdlife International 2016a).

The chinstrap penguin (*Pygoscelis antarcticus*) is one of three living *Pygoscelis* species and one of the most important top predators in the Antarctic Peninsular region (Kokubun et al. 2015). It has a circumpolar distribution, and its diet is based on Antarctic krill, *Euphausia superba* (Polito et al. 2016). During recent decades, its populations declined due to both climate-driven and fisheries-related changes in the availability of krill (Kokubun et al. 2015). However, it is currently classified as of Least Concern by the IUCN (Birdlife International 2016b).

Given their geographic distributions and position in their respective food webs, Humboldt and chinstrap penguins are ideal indicator species for assessing environmental Hg levels. Specifically, we expect to find differences in Hg levels between these penguin species, due to differences in diet and trophic position, with greater values in Humboldt penguins due to its piscivory. Likewise, based on the egg depuration hypothesis, we predict that males will exhibit greater mercury concentrations than egg-laying females during the breeding season. Finally, accounting for the major Hg anthropogenic sources located in northern Chile, which generally decrease with increasing latitude, it is likely that northern Humboldt penguins will exhibit greater Hg

concentrations compared with populations of central Chile and compared with Antarctic chinstrap penguins.

To test our predictions, we analyzed blood and feather Hg concentrations in *Spheniscus humboldti* adults and feather Hg levels in *Pygoscelis antarcticus* from different reproductive colonies along the Chilean and Antarctic coasts. At all these sites, adjacent urbanization and industrialization, as well as mercury pollution sources, decreased in number as latitude increased.

Materials and methods

Study Area

The study area included three Humboldt penguin breeding colonies located in northern and central Chile and one chinstrap penguin breeding colony in Antarctica (Fig. 1). Pan de Azúcar Island (26°09'S; 70°41'W) is within the territory of the Pan de Azúcar National Park (northern Chile), 1.2 km from the coast, and is a major Humboldt penguin colony in Chile, with ~5000 individuals (Simeone et al. 2003; Herling et al. 2005; Vianna et al. 2014; Celis et al. 2014). The island is located 16 km from Caleta Palito, a site where copper tailings have been dumped directly into the sea via the Salado River for more than 70 years, releasing extremely high amounts of mercury, among other metals (Castilla and Nealler 1978; Lee and Correa 2005; Universidad de Chile 2016). In addition, despite its closeness to the National Park, the Salado River also transports untreated urban wastewater

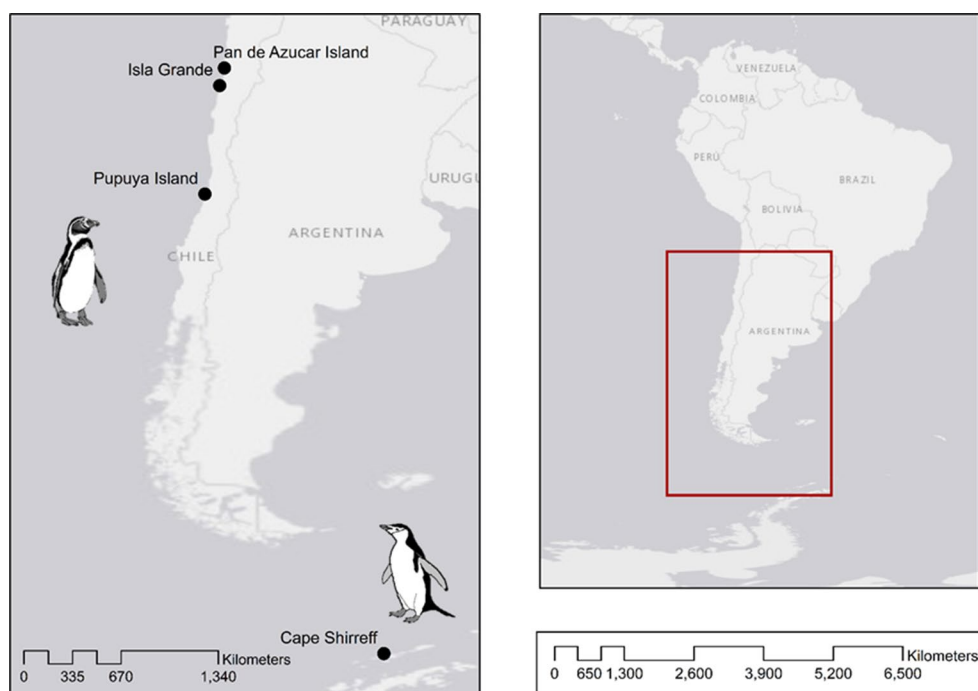
(Álvarez-Varas, personal observation). Isla Grande de Atacama (27°14'S; 70°58'W) is located 1.3 km from the coast and 120 km south of Pan de Azúcar Island and has a colony of between 30 and 60 individuals (Vianna et al. 2014). Although it is found in a gold production area, human activities impacting the area are very limited, due to a lack of nearby urban centers (Gaymer et al. 2008). This area also was decreed to be a Marine and Coastal Protected Area in 2004 (*Área Marina y Costera Protegida de Múltiples Usos* AMCP-MU in Spanish) due to its importance in conserving the Humboldt Current System and northern Chile biodiversity (Gaymer et al. 2008). Finally, Pupuya Island (33°58'S; 71°54'W) is located 1 km from the coast, facing the small rural town of Matanzas, which lacks industrial development; the closest urban center is the city of San Antonio, roughly 80 km away. The population size of Humboldt penguins there is estimated to be fewer than 20 individuals (Vianna et al. 2014).

The study site for chinstrap penguins was on Cape Shirreff, Livingston Island, in the South Shetland Islands of Antarctica (62°28'S; 60°74'W) (Fig. 1). Cape Shirreff is a frequently visited protected area and lacks industrial activities and urban centers. It supports a breeding colony of up to 10,000 pairs of chinstrap penguins (Espejo et al. 2014).

Capture and Sampling

In north and central Chile, Humboldt penguins molt in February and have reproductive peaks in spring (August–January) and autumn (April–June) (Simeone et al. 2002; Birdlife

Fig. 1 Map of the study areas in Chile and Antarctica for Humboldt and chinstrap penguins sampled in 2012



International 2016a). In this study, we collected samples from adult individuals in 2012, during the breeding season. In that year, reproductive peaks were delayed, and sampling was adjusted to classify females according to the presence/absence of eggs in nests and to test our egg depuration hypothesis. Penguins were quietly approached; a 1.5-m-long noose pole was used to lead the penguins out of their nests, and they were then captured manually. The presence of eggs in the nest of each individual was recorded (Supplementary Table). Pupuya Island penguins ($n = 14$) were sampled in January 2012 during their spring reproductive event. Penguins from Pan de Azúcar Island ($n = 24$) and Isla Grande ($n = 14$) were sampled between July and August 2012, during the autumn event. For chinstrap penguins, feathers from 16 adult individuals were collected in January 2012.

In all cases, 3 ml of blood was collected by venipuncture from either the medial metatarsal vein or the brachial vein and was stored in sodium heparin tubes at 4 °C. Samples were then frozen at -196 °C in liquid nitrogen. Feathers were manually extracted from the pectoral area and were stored in plastic bags at room temperature (Scheifler et al. 2005; Brasso et al. 2015). To avoid resampling, penguins were temporarily marked with water-resistant color markers. Captures on the Chilean coast were authorized by the Chilean Fisheries Subsecretariat (Subsecretaría de Pesca, SUB-PESCA) through Research Capture Permit #110 granted on January 16, 2012. Captures in Antarctica were performed in accordance to the Protocol on Environmental Protection to the Antarctic Treaty, and the regulation from the Scientific Committee on Antarctic Research (SCAR) provided by the Chilean Antarctic Authorities (INACH Permit 44/2012). Additionally, a bioethics permit was provided by the Pontificia Universidad Católica de Chile, following the bioethics guidelines of the Chilean National Scientific and Technological Research Commission (CONICYT).

Sex Determination

Sex in Humboldt penguins was determined through morphometric features, using discriminant function, following the protocols of Zavalaga and Paredes (1997). Two measurements were recorded: head width and bill length. All morphological measurements were determined using a Mitutoyo Digimatic Caliper with 0.01-mm accuracy, and weight was recorded using a Pesola Woltkraft HS-30 scale with of 0.1-g accuracy.

Blood Mercury Analysis

Total mercury was determined in all blood and feather samples. Blood samples were analyzed with a Direct Mercury Analyzer (DMA-80) using the Thermal-Decomposition-Amalgamation/Absorption Atomic

Spectrophotometer (TDA/AAS) technique (US-EPA 1998). The methodological precision and accuracy were tested using Certified Reference Materials (CRM) from the National Research Council of Canada (NRC-CNRC; DORM-3 fish muscle = 0.382 ± 0.06 mg/kg) and replicate samples. Blood samples were analyzed at the Heavy Metal Laboratory (DECOL), Department of Ecology, Pontificia Universidad Católica de Chile. Blood concentrations are expressed as $\mu\text{g/g}$ wet weight (w.w.).

Feather Mercury Analysis

All laboratory material was decontaminated for 2 days with HNO_3 (20%) and mili-Q water before analysis (Lopez et al. 2014). Feather samples were thoroughly washed: first with deionized water, then with acetone, and finally with 2% nitric acid, following the protocol of Cardiel et al. (2011). After washing, feathers were digested with 65% HNO_3 using a microwave system (Scheifler et al. 2005; Lopez et al. 2014) and analyzed with a Shimadzu AA-6200 atomic absorption spectrophotometer (AAS). Mercury was analyzed with a hydride vapor system HV-1 (cold vapor technique). The AAS was calibrated using custom grade standards, with detection limits of $0.007 \mu\text{g/g}$ for mercury (Centipur Merck 1000 mg/l). Further quality control included periodic blind analyses of an aliquot from a large sample of known concentration and blind runs of duplicate samples ($\pm 15\%$) during the analysis (Lopez et al. 2014). Feather samples were analyzed at the Heavy Metal Laboratory, Chemistry Department, Universidad Andrés Bello. Feather concentrations are expressed as $\mu\text{g/g}$ dry weight (d.w.).

Data Analysis

A Shapiro–Wilk’s test determined normality, and the Kruskal–Wallis nonparametric test checked for significant differences between variables. To test our predictions, we evaluated differences between species, sexes, females “with” and “without” eggs, and localities. In the same way, we determined differences in Hg concentrations among Humboldt penguin colonies through a Nemenyi’s post hoc test. A Spearman correlation was used to evaluate associations between Hg concentrations in blood and feathers and Hg levels with penguin weight (for each species and for pooled species). All analyses were performed using the statistical program R (v. 3.3.1) and the “agricolae” (de Mendiburu 2016) and “PMCMR” (Pohlert 2014) packages. Geometric means for all variables are reported as 95% confidence intervals (CI), and statistical significance was defined as $p < 0.05$.

Results

Sex Determination in Humboldt Penguins

Among the total sample of Humboldt penguins ($n=52$), 28 individuals were classified as males and 24 as females. Of the females, 17 had eggs in their nests and 7 did not. Measurements for Humboldt penguins, including sex and presence/absence of eggs, are shown in the Supplementary Table.

Mercury Concentrations in Humboldt and Chinstrap Penguins

Our results showed significant differences in Hg concentrations in feathers between species (blood was not sampled in *Pygoscelis*; $p=0.024$; Fig. 2). Mercury concentrations in blood and feathers in Humboldt penguins were 0.59 (0.47–0.74) and 2.41 (2.17–2.68) $\mu\text{g/g}$, respectively (Fig. 3A). Hg in the feathers of chinstrap penguins was 1.53 (1.49–1.57) $\mu\text{g/g}$. Mercury concentrations in each species are shown in Table 1.

Mercury Levels According to Sex in Humboldt Penguins

Comparisons of mercury levels by sex were performed using only Humboldt penguin data. For females, mean values of blood and feather Hg levels were 0.69 (0.50–0.97) and 2.44 (2.09–2.85) $\mu\text{g/g}$, respectively. For males, these values were 0.51 (0.37–0.70) and 2.39 (2.04–2.79) $\mu\text{g/g}$ respectively (Table 1). Hg levels in females with and without eggs are shown in Table 1. No significant differences between sexes were observed in blood and feathers (Fig. 3B). Similarly, in general no significant statistical differences were found when comparing blood and feather samples between males and females with and without eggs, nor when comparing females with and without eggs (data not shown). However, significant differences in blood Hg levels were seen between males

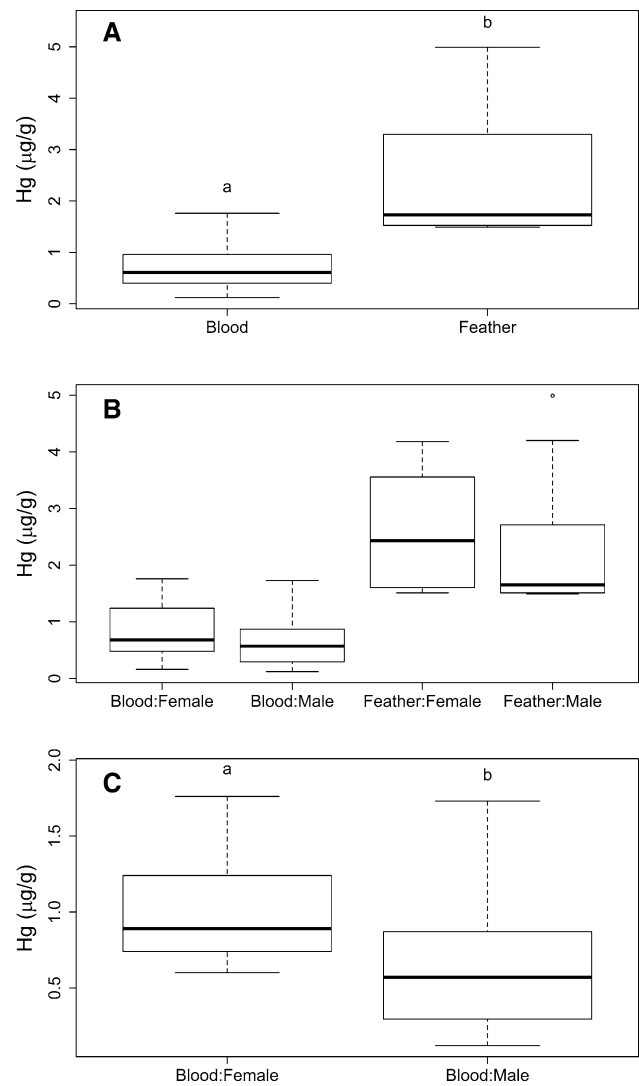


Fig. 3 Mercury concentrations ($\mu\text{g/g}$) in Humboldt penguins from Chile in 2012. **A** Hg concentrations according to tissues. **B** Hg concentrations in tissues according to sex. **C** Blood Hg concentrations in females without eggs and males. Blood and feather concentrations are expressed as $\mu\text{g/g}$ (w.w.) and $\mu\text{g/g}$ (d.w.) respectively. Different lower-case letters indicate significant differences between tissues and sexes ($p < 0.05$)

Fig. 2 Feather mercury concentration ($\mu\text{g/g}$ d.w.) for Humboldt (*Spheniscus humboldti*; SH) and chinstrap penguins (*Pygoscelis antarcticus*; PA) from Chile and Antarctica in 2012. Different lower-case letters indicate significant differences between species ($p < 0.05$)

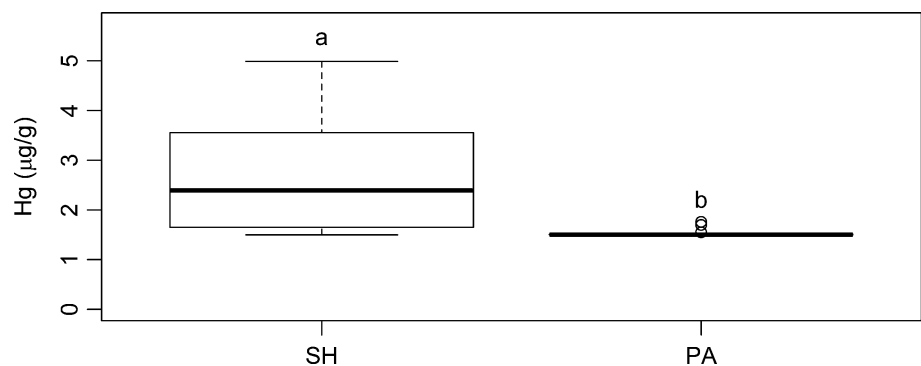


Table 1 Mercury concentrations in Humboldt and chinstrap penguins, based on sex and the presence/absence of eggs in females, at each sampled location in Chile and Antarctica in 2012

Category	Pan de Azúcar Island		Isla Grande		Pupuya Island		Cape Shirreff		Total	
	Blood	Feathers	Blood	Feathers	Blood	Feathers	Blood	Feathers	Blood	Feathers
Humboldt (<i>n</i> = 52)	0.78 (0.61–0.98)	3.08 (2.61–3.64)	0.37 (0.25–0.53)	2.39 (2.26–2.53)	–	1.59 (1.56–1.63)	–	–	0.59 (0.47–0.74)	2.41 (2.17–2.68)
Males (<i>n</i> = 28)	0.67 (0.47–0.97)	3.05 (2.34–3.95)	0.30 (0.19–0.49)	2.39 (2.11–2.71)	–	1.61 (1.55–1.66)	–	–	0.51 (0.37–0.70)	2.39 (2.04–2.79)
Females (<i>n</i> = 24)	0.92 (0.67–1.26)	3.13 (2.45–3.99)	0.45 (0.22–0.89)	2.39 (2.28–2.51)	–	1.58 (1.51–1.64)	–	–	0.69 (0.50–0.97)	2.44 (2.09–2.85)
Females with eggs (<i>n</i> = 17)	–	–	–	–	–	–	–	–	0.61 (0.39–0.94)	2.60 (2.16–3.12)
Females without eggs (<i>n</i> = 7)	–	–	–	–	–	–	–	–	0.97 (0.57–1.65)	2.09 (1.47–2.98)
Chinstrap (<i>n</i> = 16)	–	–	–	–	–	–	–	1.53 (1.49–1.57)	–	1.53 (1.49–1.57)
Males (<i>n</i> = 16)	–	–	–	–	–	–	–	1.53 (1.49–1.57)	–	1.53 (1.49–1.57)

Data shown are geometric means and 95% confidence intervals (CI; in parentheses). Blood concentrations are expressed in µg/g w.w. and feather concentrations in µg/g d.w.

and females without eggs ($p = 0.049$), showing increased concentrations in the latter (Fig. 3C).

Variation in Mercury Concentrations Between Localities

Blood mercury concentrations were compared only in Humboldt penguins from Pan de Azúcar Island and Isla Grande, because it was not possible to collect blood samples at other locations. Blood mercury values of penguins from Pan de Azúcar Island and Isla Grande were 0.78 (0.61–0.98) and 0.37 (0.25–0.53) µg/g, respectively (Table 1). Significant differences were found between both colonies ($p = 0.001$; Fig. 4A). Feather mercury concentrations in penguins at all locations are shown in Table 1. Significant differences in feather Hg levels were observed between Pan de Azúcar Island and Pupuya Island, Pan de Azúcar Island and Cape Shirreff, and between Isla Grande and Cape Shirreff (Fig. 4B).

Association Between Mercury in Blood, Feathers and Penguin Weights

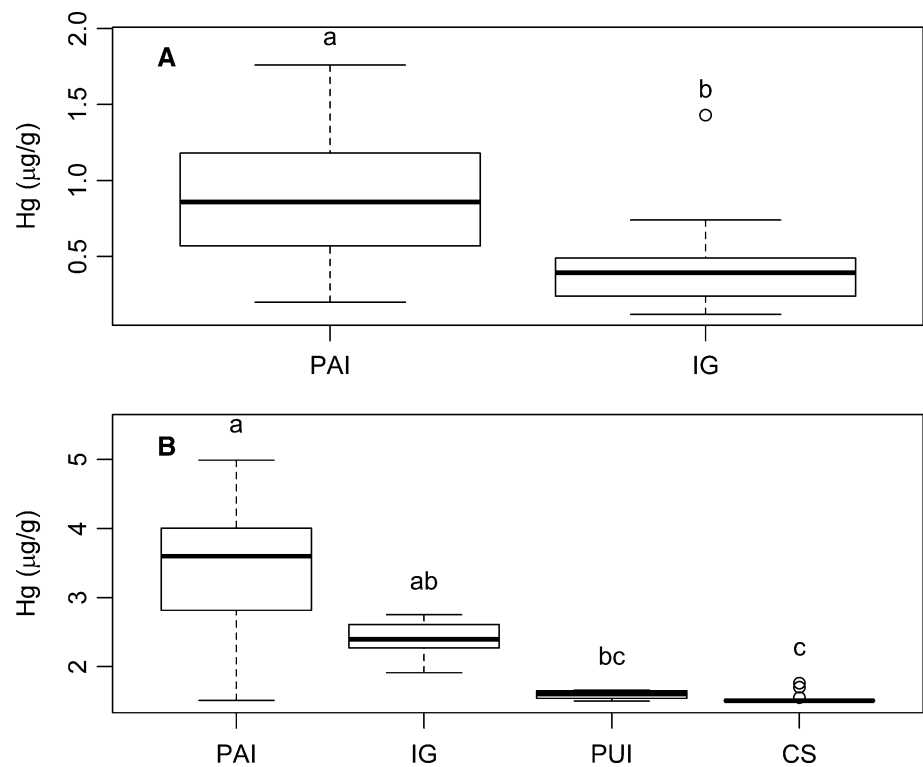
No significant relationships were observed between Hg concentrations in the blood and feathers of Humboldt penguins ($p = 0.753$). Similarly, nonsignificant results were found between weight and blood Hg levels ($p = 0.910$). However, significant negative relationships between weight and feather mercury concentration were found when considering both species (pooled species, $p = 0.041$, $R = -0.249$) and when considering female Humboldt penguins with eggs ($p = 0.044$, $R = -0.493$).

Discussion

Mercury Concentrations in Humboldt, Chinstrap, and Congener Penguins: Effects of Trophic Level

Penguins, compared with other high trophic indicator species, are suitable sentinels for local marine pollution monitoring, because they are easily sampled, wide-ranging, abundant, and long-lived, with relatively small foraging areas (Espejo et al. 2014; Finger et al. 2015). Contrary to chinstrap penguins in Antarctica, only one published study has used Humboldt penguins (and it used excreta) as indicators of heavy metal marine pollution in the Southern Hemisphere (Celis et al. 2014). Conversely, whereas different penguin species have been used in ecotoxicological studies using internal tissues and feathers, there are almost no trace metal concentration data published from the blood of these species (Table 2) (Finger et al. 2015). To our knowledge, this study

Fig. 4 Mercury concentration ($\mu\text{g/g}$) in Humboldt and chinstrap penguins at the different study sites from Chile and Antarctica in 2012. **A** Blood Hg concentration ($\mu\text{g/g}$ w.w.) in Humboldt penguins. **B** Feather Hg concentration ($\mu\text{g/g}$ d.w.) in Humboldt and chinstrap penguins. *PAI* Pan de Azúcar Island; *IG* Isla Grande; *PUI* Pupuya Island; *CS* Cape Shirreff. Different lowercase letters indicate significant differences among localities ($p < 0.05$)



represents the first investigation to report blood and feather Hg levels in Humboldt penguins.

Blood and feathers of penguins provide suitable means to investigate temporal differences of mercury contamination, and although both represent biologically relevant contaminant loads, blood analysis allows for fine-scale temporal investigations (Finger et al. 2015), because blood concentrations reflect recent dietary exposure and provide insight into seasonal variations of Hg exposure through prey (Finger et al. 2015, 2017). Mercury excretion through quills is a detoxification mechanism for birds during premolt (Becker et al. 2002). Once the boom ends its growth, the blood transport channel atrophies, leaving a permanent record until the next molt (Gochfeld and Burger 2001). Given that penguins exhibit a unique annual catastrophic molt, feather monitoring reflects storage at the time that the feather was actively growing, thus with longer annual, or intermolt, time periods (Ochoa-Acuña et al. 2002; Carravieri et al. 2013; Brasso and Polito 2013; Polito et al. 2016).

Several factors have been related to interspecific variation in metal concentrations in blood and feathers of seabirds, including feeding specialization and physiological processes of metal detoxification (Carvalho et al. 2013; Calle et al. 2015; Brasso et al. 2015). Significant differences were found in feather Hg levels between Humboldt and chinstrap penguins in this study, with greater concentrations in Humboldt penguins, as predicted. Humboldt penguins feed predominantly on pelagic fish species, particularly schooling

and shoaling species (Herling et al. 2005), such as garfish (*Scomberesox saurus*) and anchovy (*Engraulis ringens*), as well as smaller amounts of South American pilchard (*Sardinops sagax*) and Inca scad (*Trachurus murphyi*) (Wilson et al. 1989; Herling et al. 2005). In contrast, chinstrap penguins on King George Island, located in the South Shetland Islands, Antarctica, feed almost exclusively on Antarctic krill (*Euphasia superba*), far lower amounts of Antarctic silverfish (*Pleurogramma antarcticum*), and very low amounts of amphipods (Kokubun et al. 2015). These diet differences between Humboldt and chinstrap penguins demonstrate differences in their trophic levels, and therefore the Hg exposure differences detected may be due to this (Fig. 2). Hg concentrations among 10 species of penguins, representing 26 geographically distinct breeding populations in the Southern Ocean, showed that foraging at a high trophic level was associated with elevated Hg exposure. For instance, South Georgia king penguins (*Aptenodytes patagonicus*), whose diets are comprised of 97% fish, had significantly greater Hg feather concentrations than sympatric gentoo penguins (*Pygoscelis papua*), whose diets are predominantly krill, with fish comprising less than 30% (Brasso et al. 2015). Similarly, there was significantly greater Hg accumulation in gentoo penguins (which feed on krill, fish, and small squid) than chinstrap penguins (which feed exclusively on krill) at Barrientos Island, Antarctic Peninsula (Calle et al. 2015), also pointing to the implications of trophic level on mercury concentrations.

Table 2 Comparison of blood ($\mu\text{g/g}$ w.w) and feather mercury concentrations ($\mu\text{g/g}$ d.w.) between *Spheniscus* and *Pygoscelis* species in this study and previous reports

Common name	Species	Location	Blood Hg	Feather Hg	N	Source
Humboldt	<i>Spheniscus humboldti</i>	Pan de Azúcar Island, Northern Chile	0.78 ± 0.43	3.08 ± 1.03	24	This study
		Isla Grande, Northern Chile	0.37 ± 0.33	2.39 ± 0.23	14	This study
		Pupuya Island, Central Chile	–	1.59 ± 0.06	14	This study
Magellanic	<i>Spheniscus magellanicus</i>	Punta Tombo	–	0.21 ± 0.10	21	Frias et al. (2012)
		Isla Martillo, Southern Patagonia	–	1.67 ± 0.22	10	Brasso and Polito (2013)
		Staten Island, Southern Patagonia	–	2.91 ± 0.56	18	Brasso et al. (2015)
		Isla Martillo, Southern Patagonia	–	1.79 ± 0.34	16	Brasso et al. (2015)
African	<i>Spheniscus demersus</i>	Robben Island, South Africa	–	0.90 ± 0.35	10	Brasso and Polito (2013)
		Robben Island, South Africa	–	1.00 ± 0.44	19	Brasso et al. (2015)
Chinstrap	<i>Pygoscelis antarcticus</i>	King George Island, Antarctica	–	0.69 ± 0.18	10	Brasso and Polito (2013)
		King George Island, Antarctica	–	0.62 ± 0.30	16	Brasso et al. (2015)
		King George Island, Antarctica	0.20 ± 0.05	0.74 ± 0.24	20	Polito et al. (2016)
		Cape Shirreff, South Shetland Islands, Antarctica	–	1.53 ± 0.08	16	This study
		Barrientos Island, Antarctica	–	0.83 ± 0.40	33	Calle et al. (2015)
		Barrientos Island, Antarctica	–	0.24 ± 0.01	18	Calle et al. (2015)
Adelie	<i>Pygoscelis adeliae</i>	Rumpa Island, Antarctica	0.01 ± 0.01	0.17 ± 0.05	10	Honda et al. (1986)
		Terra Nova Bay, Ross Sea	–	0.82 ± 0.13	3	Bargagli et al. (1998)
		Syowa Station, East Antarctica	–	0.09 ± 0.05	10	Yamamoto et al. (1996)
		South Georgia	–	1.4	3	dos Santos et al. (2006)
		Adélie Land, Antarctica	–	0.43 ± 0.13	12	Carravieri et al. (2016)
		Adélie Land, Antarctica	–	0.66 ± 0.20	10	Carravieri et al. (2016)
		King George Island, Antarctica	–	0.32 ± 0.08	10	Brasso and Polito (2013)
		King George Island, Antarctica	–	0.35 ± 0.09	21	Brasso et al. (2015)
		King George Island, Antarctica	0.04 ± 0.01	0.35 ± 0.14	22	Polito et al. (2016)
		King George Island, Antarctica	–	0.35 ± 0.14	22	Polito et al. (2016)
Gentoo Penguin	<i>Pygoscelis papua</i>	South Georgia	–	0.95 ± 0.85	14	Becker et al. (2002)
		South Georgia	–	0.54	2	dos Santos et al. (2006)
		Estacade population, Kerguelen Island	–	4.96 ± 2.44	12	Carravieri et al. (2013)
		Crozet Island, Subantarctic Region	–	5.23 ± 1.12	12	Carravieri et al. (2016)
		Estacade population, Kerguelen Island	–	5.85 ± 3.00	12	Carravieri et al. (2013)
		Penn Island population, Kerguelen Island	–	1.44 ± 0.44	12	Carravieri et al. (2013)
		Crozet Island, Subantarctic Region	–	5.90 ± 1.91	11	Carravieri et al. (2016)
		King George Island, Antarctica	–	0.28 ± 0.05	10	Brasso and Polito (2013)
		King George Island, Antarctica	–	0.31 ± 0.10	21	Brasso et al. (2015)
		Bird Island, South Georgia	–	0.97 ± 0.67	55	Pedro et al. (2015)
		King George Island, Antarctica	0.08 ± 0.10	0.51 ± 0.51	21	Polito et al. (2016)
		Bird Island, South Georgia	–	1.13 ± 0.62	29	Pedro et al. (2015)
		Gold Harbor, South Georgia	–	0.85 ± 0.88	20	Brasso et al. (2015)
		Barrientos Island, Antarctica	–	1.83 ± 0.80	52	Calle et al. (2015)
		Barrientos Island, Antarctica	–	0.49 ± 0.20	18	Calle et al. (2015)

Data shown are mean $\pm$ 1 SD

Feather Hg concentrations in Humboldt penguins reported in this study were greater than those documented for the South Atlantic Magellanic penguins (*Spheniscus magellanicus*) and African penguins (*Spheniscus demersus*) from South Africa (Brasso et al. 2015; Table 2). In fact, the Pan de Azúcar penguins of this study showed the highest

feather Hg levels published to date for a *Spheniscus* species (Table 2). Chinstrap penguins also showed greater Hg concentrations ($1.53 \mu\text{g/g}$) compared with previous studies of this species but comparable with some subantarctic and Antarctic populations of congener gentoo (0.28 – $5.90 \mu\text{g/g}$) and Adélie penguins (0.09 – $1.40 \mu\text{g/g}$) (dos Santos et al. 2006;

Carravieri et al. 2013, 2016; Calle et al. 2015; Table 2). To date, there is no published data on blood Hg levels in *Spheniscus* species; however, the blood Hg concentrations in Humboldt penguins in this study were greater than those reported in different *Pygoscelis* species (Table 2).

Sex Effect: Intraspecific Variation of Hg Concentrations in Humboldt Penguins

Our Humboldt penguin results show no significant differences between sexes, in either blood or feathers (Fig. 3B), which is concordant with other studies of other penguin species in the Southern Hemisphere (Becker et al. 2002: macaroni penguins; Frias et al. 2012: Magellanic penguins; Brasso et al. 2015: Adélie, chinstrap, gentoo, and rockhopper penguins; Polito et al. 2016: Adélie, chinstrap, and gentoo penguins). In addition, no significant differences were found when comparing blood and feathers of females with and without eggs. Nevertheless, there was a tendency in females to show greater Hg concentrations than males, contrary to predictions (Fig. 3B). In fact, significant differences were found in blood Hg levels between males and females without eggs ($p=0.049$; Fig. 3C). A meta-analysis of 50 studies analyzing the concentration of Hg in avian tissues suggested that sex differences in Hg loads are not simply the result of egg depuration (maternal transfer to eggs), and perhaps sex differences in detoxification via eggs are important for some species but not in others (Robinson et al. 2012). Thus, the greater values of Hg may suggest that egg-laying is not an important way to counter elevated metal exposure among Humboldt penguins.

Regarding the differences in blood Hg concentration between males and females without eggs reported ($p=0.049$; Fig. 3C), a larger sample size could provide enough statistical power to clarify significance. In the same way, molecular sexing could corroborate our results. Finally, further research on diet between sexes of Humboldt penguins from our study sites is necessary to determine if variations in trophic levels translate into different Hg concentrations.

Latitudinal Variation of Penguin Mercury Levels: Local Metal Availability and Pollution Sources

Anthropogenic and natural sources of Hg may contribute to differences in concentrations within the Southern Ocean (Pedro et al. 2015). However, several biotic and abiotic factors influence local Hg bioavailability and marine food web transfer (Brasso et al. 2015; Carravieri et al. 2016). While the risk of Hg exposure for any population relates to differences in local bioavailability, oceanographic factors (such as atmospheric Hg deposition rate, upwelling activity, and local primary productivity among others) also can lead to geographic differences in prey availability altering dietary

composition and/or trophic level at the population level. Thus, Hg levels in marine species may vary among geographically distinct populations (Brasso and Polito 2013; Brasso et al. 2015).

Seabird studies suggest that vertical Hg food web transfers have a latitudinal trend in the Southern Ocean, increasing from Antarctica all the way through to subtropical waters (Blévin et al. 2013; Carravieri et al. 2014; Pedro et al. 2015). Feather Hg concentrations in adults and chicks of all seven Southern Indian Ocean penguin species showed Hg levels increasing from high to low latitudes. As predators in simple Antarctic food webs rely directly or indirectly on a few key species, there is little Hg biomagnification; in contrast, the greater complexity of subantarctic food webs leads to stronger Hg biomagnification (Carravieri et al. 2016). This intra- and interspecific latitudinal pattern has been related to feeding ecology and increased Hg concentrations in the food web due to biomagnification (Pedro et al. 2015; Carravieri et al. 2016).

The Hg concentrations reported in this study also show a latitudinal pattern; penguins breeding in the north exhibited greater Hg concentrations than those in the south, independent of the species (Fig. 4B). This geographic trend in Hg concentrations may be associated with the geographic pattern of mining activities described in the Introduction. The similarity in feather Hg levels between both northern penguin colonies (Pan de Azúcar Island and Isla Grande; Fig. 4B) could be associated with historical mining pollution, because feathers act as indicators of long-term exposure (compared with blood) (Ochoa-Acuña et al. 2002; Brasso and Polito 2013; Carravieri et al. 2016). Given that Humboldt penguins off the breeding season can travel large distances (> 150 km; Culik 2001), feather Hg concentrations would not be indicative of local pollution. Conversely, differences in blood Hg concentrations between penguins of both colonies (Fig. 4A) reflect recent dietary exposure (Finger et al. 2015), either at the time of sampling or the breeding season. Because Humboldt penguins forage 15–35 km away from the colony during the breeding season (Culik et al. 1998; Rey et al. 2013), blood Hg concentrations would reflect the degree of local pollution or metal bioavailability, which varies significantly. Thus, differences between Pan de Azúcar Island and Isla Grande could be related to local pollution sources and/or some oceanographic factors. Foraging ecology also could influence these differences (Brasso et al. 2015; Pedro et al. 2015; Carravieri et al. 2016).

However, the similarity in feather Hg concentration between central Chile (Humboldt penguins) and Antarctic penguins (chinstrap penguins; Fig. 4B) could suggest either interspecific differences in diet or mainly due to local pollution and/or Hg bioavailability. A study on trace metals (including Hg) in Humboldt penguin excreta at some important nesting sites in the northern coast of Chile (Pan de

Azúcar, Chañaral, and Cachagua Islands) showed elevated chemical levels at sites closer to pollution sources than those with no industrial presence; Pan de Azúcar Island had the highest concentrations of Hg (Celis et al. 2014), congruent with our results. Previous studies in other penguin species also have found a relationship between tissue Hg concentrations and the degree of industrialization near foraging zones. For instance, trace element concentrations (including Hg) in the blood and feathers of little penguins (*Eudyptula minor*) at three locations with varying degrees of anthropogenic contamination in South Eastern Australia provided strong evidence of a pollution gradient, with the highest Hg values in penguins near the most polluted site, and the lowest from the least contaminated location (Finger et al. 2015).

Mercury Environmental Variation and Its Impact on Penguin Population

Several studies have reported a positive correlation between blood and feather Hg concentrations in seabirds (Evers et al. 1998; Bearhop et al. 2000; Finger et al. 2015; Polito et al. 2016). A strong correlation between molt blood and feathers in little penguins suggested that blood may be a proxy for Hg sequestered into the new feathers and a relative stability of Hg in the prey of these penguins during the period before molting (Finger et al. 2015). This can occur if the molt and blood collections occur in similar temporal proximity. Our results contrast with some previous reports, because a non-significant association was observed between Hg concentration in blood and feathers of Humboldt penguins ($p=0.753$), similar to other publications (Evers et al. 1998; Polito et al. 2016). The lack of a relationship between both monitoring units reported in this study could suggest a high variability in Hg bioavailability in the study area (including in their prey). This makes sense mainly in the colonies of northern and central Chile, because they encounter high environmental heterogeneity due to ENSO events (El Niño Southern Oscillation) and upwelling activity associated with the Humboldt Current System (Thiel et al. 2007).

The negative association found between feather Hg levels and penguin weight in this study ($p=0.041$, $R=-0.249$) may be related to various adverse effects on birds, such as reduced food intake leading to weight loss, progressive weakness, decrease of fecundity, hatching, and fledging rates, and an increase of mortality rates (Ochoa-Acuña et al. 2002). On St. Kilda, Australia, the seasonal variation of metals in blood of little penguins showed a negative correlation of flipper length with blood Hg concentrations, suggesting a possible deleterious physiological effect of pollution in their foraging area (Finger et al. 2016). A study in Humboldt penguins reported that body condition and growth rates of chicks on Islotes de Puñihuil (south of Chile) were better and higher than on Pan de Azúcar Island (our study area),

which was attributed mainly to variability in food abundance and availability (Miranda 2015). Nevertheless, the greater values of Hg reported on Pan de Azúcar Island in the present study could be related to the decreased growth rate of Humboldt penguin chicks reported by Miranda (2015), reflecting a negative impact of Hg at the population level. Although Humboldt and chinstrap penguins did not exceed the commonly used feather toxicity threshold of 5 µg/g (Burger and Gochfeld 1997; Carravieri et al. 2013), the values reported require further investigation, especially on the potential negative effects of Hg on these populations. Thus, studies on reproductive performance, body condition, and growth and mortality rates are necessary to better interpret elevated Hg levels. In addition, our results highlight the need of increasing knowledge on levels of other heavy metals in penguins from highly susceptible colonies in northern Chile to mitigate potential negative impacts.

Conclusions

The two penguin species investigated in this study showed elevated blood and feather Hg concentrations, highlighting the relevance to investigate negative effects associated to this metal on these species, both at the individual and population levels. Although greater Hg burdens in Humboldt penguins than in chinstrap penguins are probably due to variation in diet and trophic level, our results suggest an important effect of the degree of Hg pollution adjacent to foraging areas. Further research on Hg content in environmental samples and penguin prey species is needed to comprehend Hg bioavailability in each location and, consequently, the importance of local Hg pollution levels. In addition, a larger overall sample size, and research into diet and trophic level, will uncover the reasons for the differences seen in Hg concentrations between sexes within Humboldt penguins. Ultimately, this study highlights the critical need for mercury monitoring in penguin populations along the coast of Chile, especially in places where mining activities continue to pose significant environmental threats.

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Compliance with Ethical Standards

Conflict of interest The authors declare that they have no conflict of interest.

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